Igor Grybek

Capital Operations, Corporate Banking & Technology Decision Systems

Private Capital | Financing Feasibility | Infrastructure | Data, AI & Risk Governance

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Education

Warsaw School of Economics (SGH) 2026 – Present
Master of Science (MSc) in Quantitative Methods in Economics and Information Systems (MIESI) Warsaw, Poland

Core Specialization: Econometric Modeling, Stochastic Processes, Machine Learning, Information Systems Architecture, Business Intelligence & Advanced Financial Engineering.

Warsaw School of Economics (SGH) 2021 – 2026
Bachelor of Finance and Accounting Warsaw, Poland

Relevant Coursework: Corporate Finance, Financial Markets, Investment Analysis, Corporate & Investment Banking, Econometrics, Mathematical Statistics, Advanced Macroeconomics, Financial Accounting.

Bachelor Thesis: *The Impact of Institutional Financing on the Operational Stability of Deep Tech Companies: A Case Study of Creotech Instruments S.A.* (Empirical examination of institutional capital structures, debt/equity sequencing, grant co-financing, liquidity buffers, and milestone execution under high capital intensity).

Professional Experience

BNP Paribas Corporate and Institutional Banking (CIB) Jun 2025 – Present
Private Capital Specialist Warsaw, Poland

- **Asset Verification & Reporting:** Led asset inventory verification, quarterly reporting, and bi-yearly asset certifications across diverse portfolio structures—funds of funds, private equity, and real estate funds with underlying SPVs—ensuring strict AIFMD compliance and operational risk minimization.
- **Portfolio & Cash Operations:** Managed daily cash monitoring, multi-entity reconciliations, and complex transaction booking for capital calls, distributions, purchases, sales, and debt/equity financing across a $150B AUM client investment ecosystem.
- **Legal Review & Title Integrity:** Audited critical ownership and governance documentation (LPAs, subscription agreements, shareholder agreements, notarial deeds, registers, and capital statements) to verify title and proactively resolve counterparty discrepancies.
- **Depositary Platforms & Automation:** Leveraged FIA eFront for advanced fund administration and depositary controls; contributed to testing and integrating proprietary RPA and AI/ML document-processing pipelines for automated operational workflows.

Arxa Group / Portfolio of Technology Projects Mar 2023 – Mar 2025
Co-Founder | R&D Project Lead | Investment & Business Associate Warsaw, Poland

- **Venture Architecture & Product Management:** Directed R&D, product architecture, and commercial strategy across four proprietary technology initiatives: EDNA (enterprise knowledge infrastructure & governed RAG), ARXA (industrial telemetry & real-options operating engine), ARXIS (AEC architecture traceability & change governance), and Hobbly (transaction state machine & services platform).
- **Financial Decision Modeling:** Built institutional financial decision models (IRR, DCF, ARR forecasts, and break-even sensitivity) connecting technical architecture, unit economics, and capital structure to evaluate financing feasibility and projected returns for capital allocation.
- **Market Intelligence & Investor Relations:** Conducted 11 in-depth quantitative and qualitative cross-sector studies (fintech, blockchain, defense, manufacturing, digital infrastructure, AI, and quantum tech); cultivated partnerships and pitched research-backed investment theses to over 10 HNI and VC/PE funds across Europe and the US ($2.3B+ AUM network).
- **Commercial Feasibility & Unit Economics:** Formulated go-to-market strategies, unit-economics models (CAC, LTV, contribution margin), and technical feasibility benchmarks, aligning product roadmaps with investor criteria and customer operational constraints.

Santander Corporate and Investment Banking (CIB) Oct 2022 – Mar 2023
Internship / Credit Analyst, Corporate Banking Coverage Warsaw, Poland

- **Deal Structuring & Documentation:** Prepared and structured credit documentation and comprehensive banking product offers for major corporate clients (minimum PLN 1.2B EBITDA across TMT, retail, manufacturing, heavy industry, automotive, and energy coverage).
- **Financial Modeling & Credit Underwriting:** Built dynamic financial models, evaluated debt-bearing capacity, analyzed liquidity profiles, and assessed operational risks to support senior coverage directors in credit underwriting and deal-structuring decisions.
- **Cross-Divisional Deal Execution:** Connected analytical and documentary inputs across relationship managers, structured/project finance, risk, and product specialists to deliver financing solutions tailored to clients' strategic capital requirements.

SDG / DGC Oct 2021 – Oct 2022
Project & Business Development Support CEE

- **Infrastructure Feasibility & Due Diligence:** Participated in development-stage renewable energy (photovoltaics, onshore wind) and infrastructure feasibility in CEE through market analysis, preliminary due diligence, site and regulatory reviews, and project documentation.
- **Stakeholder Alignment & Executive Reporting:** Coordinated executive briefings, engineering partner alignments, and stakeholder meetings, assembling critical documentary inputs to support project evaluation, sequencing, and capital commitments.

Executive Programmes & Professional Specializations

Financial Engineering, Valuation & Capital Allocation

Columbia University | *Financial Engineering & Risk Management Specialization*
Key Modules: Term-Structure Models, Credit Derivatives, Derivative Pricing, Computational Methods & Calibration, Asset Management Optimization, Dynamic Programming & Real Options.

Corporate Finance Institute (CFI) | *Financial Modeling & Valuation Analysis (FMVA)*
Key Modules: Comparable & DCF Valuation, Financial Statement Analysis (3-Statement Modeling), M&A Modeling, Credit & Risk Analysis, Capital Budgeting, Corporate Finance, Wealth Management.

Wharton School, University of Pennsylvania | *Business, Finance & Quantitative Modeling*
Key Modules: Corporate Finance, Risk & Scenario Modeling, Simulation Analysis, FinTech Foundations, AI Strategy & Digital Asset Regulations.

Università Bocconi & Erasmus University Rotterdam | *Private Equity, Infrastructure & Advanced Valuation*
Key Modules: PE/VC Deal Structuring, Infrastructure Project Financing, M&A Strategy, LBO Valuation Modeling, Corporate Restructuring.

Duke University | *Financial Management & Decentralized Finance (DeFi)*
Key Modules: Capital Investment Evaluation, Cash Flow Optimization, Energy & Commodity Markets, Decentralized Infrastructure & Financial Risk Primitives.

Interactive Brokers (IBKR) | *Capital Markets, Portfolio Management & Derivatives*
Key Modules: Multi-Asset Trading (Equities, FX, Fixed Income, Options), Value, Growth & Momentum Investment Strategies, Macroeconomic Cycle Analysis.

Systems Architecture, Data Infrastructure & Applied AI

IBM | *Systems Analysis, Cloud Architecture & Enterprise AI*
Key Modules: Enterprise Systems Design (SIPOC, RACI, DFD), Microservices & Cloud Infrastructure, Python/Flask AI Development, DevOps/CI-CD, Agile Systems Analysis.

Applied Enterprise Data Platforms | *Palantir Foundry Architecture & Data Intelligence*
Key Modules: Platform Architecture, AIP Integration, Data Pipelines & Connectors, Dynamic Ontologies, Quiver Quantitative Analytics, Contour Data Exploration.

Johns Hopkins University | *Advanced Statistics & Data Science*
Key Modules: Mathematical Statistics, Regression Inference, Predictive Analytics, High-Dimensional Data Processing, Stochastic Simulation.

Stanford University | *Machine Learning Algorithms & Product Strategy*
Key Modules: Supervised & Unsupervised Learning, Reinforcement Learning, Algorithmic Optimization, Go-to-Market Strategy & Product Leadership.

Microsoft | *Enterprise Data Analytics & Power BI*
Key Modules: Advanced DAX Modeling, Power Query ETL, Lakehouse Architecture, Interactive Executive Reporting Dashboards.

Technology Architecture & Quantitative Decision Platforms

EDNA (Governed Knowledge Infrastructure & Deterministic Provenance): Architectural Python reference pipeline applying GxP/FDA Part 11 and EU AI Act data integrity frameworks (IBM/JHU); pairs multi-hop semantic retrieval with cryptographic SHA-256 chunk lineage and deterministic contradiction resolution to eliminate AI hallucinations in high-stakes corporate credit and clinical underwriting.

ARXA (Industrial Telemetry & Real-Options Decision Engine): Physical-financial asset model in Python linking high-frequency vibration/thermal degradation curves with Columbia-modeled binomial real-options lattices (CRR); transforms continuous Remaining Useful Life (RUL) telemetry into bankable CapEx vs. OpEx maintenance timing and energy-curtailment decisions.

ARXIS (Industrial Dependency & Change-Management DAG Platform): Directed Acyclic Graph (DAG) platform in Python incorporating IBM Systems Analysis (SIPOC/RACI) to eliminate multi-disciplinary engineering clash rework; maps critical-path CPM dependencies across architectural, MEP, and structural lifecycles to enforce auditable change-order governance.

Hobbly (Transaction Escrow State Machine & Settlement Rails): Deterministic finite state automaton (FSM) in Python/SQLite governing multi-tier escrow lifecycles under PSD2/SCA constraints; enforces automated reconciliation, strict transaction idempotency, and state-locked dispute arbitration to systematically protect marketplace contribution margins.

Capabilities & Technical Tooling

Financial & Valuation Modeling: 3-Statement Modeling, DCF, LBO, M&A Comparables, Real Options Lattices, Debt Sizing, Scenario Sensitivity, Liquidity Stress Testing, AIFMD Fund Controls.

Data, AI & Enterprise Platforms: Python (NumPy, Pandas, Scikit-learn), SQL, Power BI (Advanced DAX, Power Query), Palantir Foundry, FIA eFront (Fund Operations & Custody Reconciliation), MATLAB, LaTeX, Git, Jupyter.

Venture & Systems Tooling: Jira, Linear, Figma, Notion, EMIS Intelligence, Cursor, IBKR Trader Workstation • **Languages:** Polish (Native), English (C1/C2 Proficient, Cambridge CAE score 197, Pass Grade B).

Honors, Competitions & Professional Affiliations

Honors & Competitions: Finalist, BCG Platinum Technology Consulting Challenge 2026; Laureate, Artificial Intelligence (AI4US) Competition (Ministry of Development and Technology); Participant, National Physics Olympiad (analytical problem solving).

Networks & Communities: Warsaw Quantum Computing Group, Cobin Angels, Quantum AI Foundation, AI Tinkerers Warsaw • **Executive Workshops:** Kozminski PE Masterclass, SGH Banking and Startup Society.